

AI BASED REAL TIME POSTURE DETECTION

Pratyasha Singh

*School of Computing Science Engineering
and Artificial Engineering*

VIT Bhopal

Bhopal, India

pratyasha.25mip10119@vitbhopal.ac.in

Snehal Dixit

*School of Computing Science Engineering
and Artificial Engineering*

VIT Bhopal

Bhopal, India

snehal.mip10072@vitbhopal.ac.in

Abstract—Prolonged sitting and improper body alignment can contribute to physical discomfort and poor ergonomic habits. This paper presents a browser-based AI-assisted posture monitoring system that uses MediaPipe Pose and computer vision to provide real-time feedback on upper-body posture. The system processes camera frames locally to extract relevant facial and shoulder landmarks, including the nose, eyes, ears, and shoulders. Two normalized geometric features are calculated to estimate posture asymmetry and the relative distance between the face and shoulders. These measurements are compared with a calibrated upright-posture baseline to identify three posture conditions: upright posture, slouching, and lateral leaning. The application provides visual feedback, session tracking, and event counters to improve users' awareness of their sitting posture. A landmark-visibility and geometric-validity verification stage is also incorporated to reduce unreliable predictions caused by incomplete or implausible upper-body detections. Since the system operates in the browser without transmitting camera frames to an application server, it offers a privacy-conscious and accessible approach to posture monitoring. The proposed system is intended as a posture-awareness aid rather than a medical diagnostic tool.

Index Terms—MediaPipe Pose, Computer Vision, Posture Monitoring, Pose Estimation, Geometric Features, Calibration, Real-Time Feedback, Human-Computer Interaction

I. INTRODUCTION

Maintaining proper body posture is an important aspect of physical well-being, particularly for individuals who spend extended periods sitting while studying, working, or using digital devices. Prolonged sitting in an improper posture may contribute to discomfort, fatigue, and musculoskeletal strain. However, continuous manual monitoring of posture is inconvenient and difficult to maintain in everyday environments. This creates a need for accessible, non-invasive, and real-time posture monitoring systems.

Recent developments in artificial intelligence and computer vision have made it possible to analyze human body posture using ordinary cameras. Pose estimation techniques identify important body landmarks and provide geometric information about the position and alignment of different body parts. Compared with wearable-based monitoring systems, camera-based approaches can offer a contactless solution without requiring users to attach sensors to their bodies.

This paper presents a browser-based posture monitoring system that uses MediaPipe Pose to estimate upper-body landmarks from camera input. The system evaluates posture

using normalized geometric features derived from facial and shoulder landmarks. One feature measures left-right asymmetry between face-to-shoulder distances, while another evaluates the combined face-to-shoulder distance relative to shoulder width. These measurements are compared with a calibrated upright-posture baseline to classify the user's posture.

The proposed system provides feedback for three posture states: upright, slouching, and lateral leaning. It also includes a calibration mechanism that adapts the baseline to the user's body proportions and camera position. To improve reliability, geometric validation checks are applied to reject incomplete or implausible upper-body detections. The application displays posture status, session duration, and posture-event counters through a responsive web interface.

Since the system processes camera frames within the browser and does not transmit or store those frames, it is designed as a privacy-conscious posture-awareness tool. The proposed approach is intended for educational, office, and home environments where users can receive immediate awareness of posture deviations. However, the system is a heuristic posture aid and is not intended to diagnose or treat medical conditions.

II. PROPOSED METHODOLOGY

The proposed system is a camera-based posture monitoring application that uses computer vision to analyze the user's body posture. The system captures video through a camera, processes the input using a pose estimation framework, extracts relevant body landmarks, and evaluates posture using geometric relationships. A calibration mechanism establishes a reference posture, which is used to identify deviations during monitoring.

The complete methodology consists of camera input acquisition, pose landmark detection, feature extraction, reference calibration, posture classification, and feedback generation. The system is designed to operate through a browser-based interface, allowing users to monitor their posture without requiring wearable sensors.

A. System Architecture

The system architecture consists of five major components: camera input, pose estimation, geometric feature extraction, posture assessment, and user feedback. The camera captures

the user's upper-body posture. The pose estimation module identifies relevant landmarks, which are passed to the feature extraction stage. The extracted measurements are compared with a calibrated reference to determine the posture status. The result is then displayed through the user interface.

The overall workflow is represented as follows:

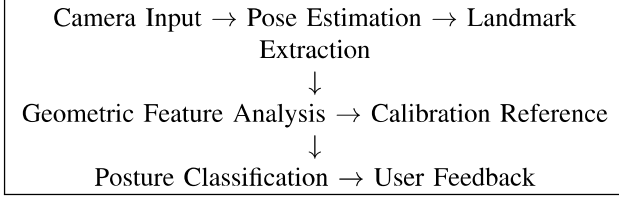


Fig. 1. Workflow of the proposed posture monitoring system.

B. Pose Estimation

The system uses MediaPipe Pose to identify body landmarks from the camera input. Pose estimation provides the spatial coordinates of relevant body points, which can be used to analyze body alignment. The detected landmarks form the foundation of the posture assessment process.

The system focuses on upper-body landmarks relevant to posture analysis. The quality of posture assessment depends on the visibility and reliability of these landmarks. When the required landmarks are not detected reliably, the system can avoid making an invalid posture assessment.

C. Geometric Feature Extraction

After detecting the body landmarks, the system calculates geometric relationships between selected landmarks to represent the user's posture. These relationships provide information about body alignment and deviations from the reference position.

The extracted features are based on the relative positions of facial and shoulder landmarks. Using relative or normalized measurements helps reduce the influence of changes in image scale and the user's distance from the camera. The resulting features are used by the posture assessment logic to identify possible slouching or lateral leaning.

D. Calibration

The system includes a calibration stage to establish a reference for the user's upright posture. During calibration, the user is expected to sit in the desired upright position. The system records posture-related measurements from this reference position.

The calibrated values are subsequently used to compare the current posture with the expected posture. This approach allows the system to account for differences in user body proportions and camera placement. However, the calibration is dependent on the user maintaining a suitable reference posture and keeping the camera position consistent.

E. Posture Classification

The posture classification stage compares the current geometric features with the calibrated reference values. Based on the implemented decision rules, the system determines whether the observed posture is within the expected range or indicates a posture deviation.

The system is intended to identify posture-related deviations such as slouching and lateral leaning. The classification is based on geometric analysis rather than medical assessment. Factors such as camera angle, lighting, body occlusion, and incorrect landmark detection may influence the classification outcome.

F. Feedback and Monitoring

The system presents the detected posture status through a user interface. This feedback is intended to improve the user's awareness of posture deviations and encourage the maintenance of an upright sitting position.

The monitoring interface may display posture status and session-related information depending on the implemented features. By providing feedback during camera-based monitoring, the system aims to support users in developing better posture awareness during prolonged sitting activities.

III. IMPLEMENTATION

The proposed system is implemented as a browser-based application using web technologies and a computer vision pose estimation framework. The application integrates camera access, pose landmark detection, geometric feature processing, posture assessment, and a graphical user interface.

The implementation is organized into functional components that handle camera input, landmark processing, posture evaluation, and user-interface updates. This organization allows the posture assessment logic to operate separately from the presentation layer and makes the system easier to maintain and extend.

TABLE I
TECHNOLOGIES USED IN THE PROPOSED SYSTEM

Component	Technology
Pose Estimation	MediaPipe Pose
Programming	JavaScript / Project Implementation
Interface	HTML and CSS
Input	Camera Feed
Processing	Geometric Feature Analysis

IV. EXPERIMENTAL SETUP

The system can be evaluated using camera-based posture samples representing different sitting conditions. The evaluation should include an upright posture and intentionally altered postures, such as forward slouching and lateral leaning, where these categories are supported by the implementation.

During testing, the system should be observed under different conditions, including changes in camera position, lighting, and user posture. The evaluation should record the predicted

posture, the actual posture category, and the response of the feedback mechanism.

The following parameters may be recorded during evaluation:

- Correct and incorrect posture classifications.
- Detection response under different posture conditions.
- Consistency of predictions during continuous monitoring.
- Response time of the feedback mechanism.
- Effect of camera position and lighting on detection.

V. RESULTS AND DISCUSSION

The proposed system was examined to understand its ability to identify posture-related deviations using camera-based pose estimation. The evaluation focused on the functioning of the camera input, landmark detection, geometric feature analysis, posture classification, and feedback display.

The system is expected to identify an upright posture and detect deviations such as slouching and lateral leaning when the relevant landmarks are visible and the camera is positioned appropriately. The calibration mechanism provides a reference for comparing subsequent posture observations.

The final performance of the system should be reported using actual experimental observations. Suitable evaluation measures include classification accuracy, precision, recall, F1-score, detection consistency, and response time, depending on the testing methodology. These values should be calculated from recorded test results rather than estimated.

The system's practical performance may be affected by camera angle, lighting, body occlusion, subject distance, and variations in sitting posture. These factors should be considered when interpreting the experimental results.

TABLE II
POSTURE CLASSIFICATION RESULTS

Actual / Predicted	Upright	Slouching	Leaning
Upright	–	–	–
Slouching	–	–	–
Leaning	–	–	–

VI. LIMITATIONS

The proposed system has several limitations. Its performance depends on the quality of camera input and the visibility of the body landmarks required for posture analysis. Changes in camera position, lighting conditions, subject distance, and partial body occlusion may affect the detected landmarks and classification results.

The posture assessment is based on geometric rules and a calibrated reference. Therefore, it may not represent every possible posture variation or individual ergonomic requirement. The system is intended for posture awareness and is not a medical diagnostic or treatment tool.

Further evaluation with a larger and more diverse set of users and posture samples is required to assess the generalizability and robustness of the approach.

VII. CONCLUSION AND FUTURE SCOPE

This paper presented an AI-based posture monitoring system that uses computer vision and pose estimation to assess posture through camera input. By extracting body landmarks, analyzing geometric relationships, and comparing the observed posture with a calibrated reference, the system provides a non-invasive approach to posture awareness.

The browser-based design allows the system to be used with a standard camera without requiring wearable sensors. The proposed approach can support users in becoming more aware of posture deviations during prolonged sitting activities.

Future work may include improving the robustness of posture classification, evaluating the system using larger and more diverse datasets, incorporating temporal analysis to reduce unstable predictions, and supporting additional posture categories. Further improvements may also include enhanced feedback, personalized thresholds, and detailed posture-monitoring reports.

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